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	Departamento Department	Stress Methods & Opt. (TAESP2)	Avión Aircraft	Technologies
<u>Título/Title</u> Program URCO Version 5 User's Manual				
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<u>Resumen/Summary</u> The present document includes the user's manual of program URCO, version 5. This program calculates the allowable values for fastener joints made in composite material for Groups III, IV and V, tape and fabric, according to the official Airbus Defence & Space general procedure. It also calculates the reserve factors of the joints for several sets of loads introduced by the user. This report also includes criteria for the determination of the geometrical parameters defining the joint. Allowable design values in URCO take into account material exposure to airframes typical ambient design conditions, as these allowable values are supported by tests with coupons moisture saturated in ambient at 70°C and 80%RH. These aged coupons were tested at 70°C. The allowable values and static margins provided by this tool are not valid for strength analyses of joints when the local temperature exceeds 70°C. PROPIEDAD DE AIRBUS DEFENCE AND SPACE S.A.U. Este documento no puede ser utilizado ni reproducido total o parcialmente sin la previa autorización escrita de la Dirección de Ingeniería de Aviones Militares AIRBUS DEFENCE AND SPACE S.A.U.. AIRBUS DEFENCE AND SPACE S.A.U. PROPERTY. This document shall neither be used nor completely or partially reproduced without previous written authorization of AIRBUS DEFENCE AND SPACE S.A.U. Military Aircraft Engineering Directorate.				
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1 INTRODUCTION

1.1 TECHNICAL SHEET OF THE PROGRAM

Technical Report: ME-T-NT-100027

PROGRAM

URCO

DATE

Apr/2017

OBJECTIVE

This program calculates the allowable values for fastened joints made in composite as well as the reserve factors for a set of loads, up to 70°C/W conditions.

LANGUAGE

Fortran 77

Visual Basic Net

DEVELOPMENT TOOLS

Oracle Solaris Studio

MS Visual Studio 2013

OS/PLATFORM

Unix (Solaris), Linux (Red Hat/Intel) & Win (x86)

Windows (x86)

NUMBER OF LINES (APPROX.)

3300 (Fortran, analysis core)

+2100 (Visual Basic .Net)

Total 5400

EXECUTION MODE

Console (input data in text file)

Interactive GUI

Notes:

- Blank lines and commented lines excluded.
- Count for Visual Basic code excludes lines created by Visual Studio in designer.vb files.

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VERSION

5.0

5.2

5.3

5.4

ADDITIONAL DOCUMENTS

See list of applicable references in sub-chapter 1.5.

1.2 SOFTWARE PURPOSE AND DESCRIPTION

The present document includes the User's Manual for version 5 of the URCO Program, which has been developed by EADS CASA since year 1999.

This program calculates the allowable values for fastened joints made in composite material for Groups III, IV and V, tape and fabric, according to the Military Transport / Airbus Defence & Space (former EADS CASA) official procedure.

These allowable values are the following:

- **A:** Tension or compression strength of the laminate, in the gross cross section located behind fastener hole, only submitted to pure by-pass load. It is provided in terms of microstrains ($\mu\epsilon$).
- **B:** Tension strength of the laminate, in the net cross section located at the fastener hole, only submitted to the fastener shear load. It is provided in terms of bearing stress units (e.g. MPa).
- **C:** Bearing strength of the laminate at the fastener hole, only submitted to the fastener shear load. It is provided in terms of bearing stress units (e.g. MPa) and failure is assumed once the maximum deformation of the hole is equal or higher than 6% of its diameter.

Allowable design values in URCO take into account material exposure to airframes typical ambient design conditions, as these allowable values are supported by tests with coupons moisture saturated in ambient at 70°C and 80%RH. These aged coupons were tested at 70°C. The allowable values and static margins provided by this tool are not valid for strength analyses of joints when the local temperature exceeds 70°C.

The program also calculates the reserve factors of the joint for several sets of loads introduced by the user. These reserve factors are the following ones:

- Reserve Factor for laminate failure in the net cross section located at the fastener hole due to tension or compression.
- Reserve Factor for laminate bearing failure at the fastener hole.

URCO program database tests were performed with threaded fasteners installed with precision fit Type C as per Ref. 4 or equivalent.

Several analysis guidelines for additional applications of URCO program output data, in terms of solving typical design configurations outside the standard use of the software, are provided in Appendix A, such as mixed laminates joints, installed repair bushings, multi-axially loaded laminates or joints with non-structural shimming analyses.

1.3 DIFFERENCES AMONG VERSIONS

Version 1.0:

Initial version developed in Fortran 77.

In parallel with this version, a Windows application was issued for interactive input and output. This application works interactively with the user. The analysis is performed by means of a call to the

original FORTRAN program. The data interchange is done through temporary files, deleted at the end of the execution.

Version 2.0:

The Graphic User Interface (GUI) is modified, reducing its size and including figures to clarify the input and output data meaning.

The user can choose the language: Spanish or English.

The program is improved to allow the creation of a dynamic link library DLL for Windows operative system.

Version 3.0:

The FORTRAN version is fully rearranged.

The interpolation method in the internal data is modified. In previous versions, the interpolation was done by 'splines', calling to routines included in the IMSL library. In this version, a linear interpolation is performed. This fact eliminates the dependency from the IMSL software.

The internal data is extrapolated, avoiding some functional limitations of the program and advising with appropriate warnings to the user when the considered data is extrapolated.

The Graphic User Interface includes now the possibility to directly send the problem results to a printer.

Version 3.1:

The program version is explicitly declared at the bottom of the output file.

Version 3.2:

Output data file drawing improved in order to avoid mistakes with backslashes in Multivac console.

Version 4.0:

The internal data of the program corresponding to tape of Group V material are modified according to Ref. 3.

Version 4.1:

The way to calculate the effective distance to the edge has been modified. In previous version was calculated as:

- Minimum value of: Distance to the plate edge, measured in the load direction. Fastener pitch, measured in the load direction.

$$e_{\text{eff}} = \min(e, p)$$

Now, it is calculated as:

- Minimum value of: Distance to the plate edge, measured in the load direction. Fastener pitch less a hole radius, measured in the load direction.

$$e_{\text{eff}} = \min(e, p - D/2)$$

Version 4.2:

The program arrangement is slightly improved. The changes do not affect to the calculation results.

The Graphic User Interface includes the possibility to show a list with several references to additional test results and results for other materials not included in the program main database.

Version 5.0:

The data flow of the program is modified. Now, the input data as well as the output results is performed by means of the following files:

- Input file: With the input data grouped in the same way as for other applications.
- Output file: With all the input data and output results in printable form.
- Numeric file: Containing only numeric results. This file can be useful as input file for other applications.

The reserve factors for a fixed geometry can be calculated for several load cases using the current program version. In previous version, only one load case was allowed.

In the same way as for the Visual Basic version, in the official FORTRAN version it is possible to select different units for forces and lengths:

- Force: N kg lb
- Length: mm cm in

Version 5.1:

Program adapted to environment Visual Studio 2010 / Intel Parallel Studio XE 2013.

Version 5.2:

Several visual bugs at GUI interface and output data are corrected.

Program adapted to environment Visual Studio 2013 / Visual Basic .Net Framework 4 Client Profile / Intel Visual Fortran.

Console version of program also compiled for Linux/Multivac platform.

Version 5.3:

A bug related to warning messages is corrected since no appropriate message was shown if t/D geometrical limits were exceed for single shear/countersunk joint configurations. In addition, the inadequate related warning message shown for double shear/protruding joint configurations is also corrected.

Several improvements in GUI forms' labels and language corrections in input data text file generation are added.

Program adapted to environment Visual Basic .Net Framework 4.5 Client Profile.

Other minor improvements are incorporated such as Unix/Linux consoles compatibility with input files in PC format, memory of active folder, regional and decimal local configurations, writing of header with basic information of the program and program version at results output reports, active file name at the title of the GUI window forms, English as default language, splash window adapted to new requirements and visualization compatibility with fonts enlarged to 125% and 150%.

Version 5.4:

A bug related to bearing reserve factor calculation is corrected for load cases without applied rivet shear load.

1.4 ABBREVIATIONS AND SYMBOLS

<u>Symbol</u>	<u>Units</u>	<u>Definition</u>
DLL		Dynamic-Link Library
GUI		Graphical User Interface
I/O		Input and Output
KDF		Knock-down Factor
MSVS		Microsoft® Visual Studio® 2013 (plus Intel® Visual Fortran Composer version 14 and IMSL Fortran Numerical Library version 6.0.0)
N/A		Not applicable
OS		Operating System
TTH		Tension Through the Hole
URCO		'Uniones Remachadas en Compuesto'

1.5 REFERENCES

	<u>Document</u>	<u>Issue</u>	<u>Title</u>
Ref. 1	NT-T-ADP-06004	A	'Theoretical Manual of the URCO Program, version 4' Technical Report. [Original in Spanish]
Ref. 2	NT-T-ADP-99021	A	'Programmer Manual of the URCO Program, version 1' Technical Report. [Original in Spanish]
Ref. 3	NT-0-ADP-04001	A	'Revisión del procedimiento de cálculo de uniones remachadas para cinta del Grupo V' Technical Report. [Original in Spanish]
Ref. 4	CAN40027	10	'Holes for joint elements not metric' Airbus DS standard.

2 PROGRAM EXECUTION

2.1 WINDOWS/PC VERSIONS

An application has been developed in Visual Basic for Windows/PC environment. The original routines of the program for input data, calculation and output results have been included in a DLL, which is used by this application. Therefore, the same results will be obtained, with only differences due to numerical rounding.

2.1.1 WINDOWS/PC GUI VERSION

The Visual Basic GUI version executable program `Urco.exe` (as well as an electronic copy of this manual and the program examples) is located in the folder:

```
\\Cae\InfoCalculo\Soft\Urco
```

To execute the program, the user has to click twice on the Visual Basic executable program.

2.1.2 WINDOWS/PC CONSOLE VERSION

Prior to program execution, it is necessary to create the input data file, as described in section §3.2.

The Fortran program in Windows console version is also available and its executable program `Urco.exe` (as well as an electronic copy of this manual and the program examples) is located in the folder:

```
\\Cae\InfoCalculo\Soft\Urco\Consola
```

The program asks for the language to use and the name of the input and output files, as shown in the following example:

```
Idioma / Language:
=====
1 ... ESPAÑOL
2 ... ENGLISH
Idioma / Language: 2
Name of the input file: Example.dat
Name of the output file: Example.res
Name of the numeric file: Example.num
Urco 5.4. (c) Airbus Defence & Space, 2017
```

The CPU time for calculation depends on the quantity of load cases but it is, in general, negligible.

This console application can be also executed in one single command by passing as arguments the input data file (i.e. filein.dat), the desired name of the output results file (i.e. fileout.res) and numeric file (i.e. filenum.num), and an integer number in order to select the report language (1 is Spanish, 2 is English):

```
urco.exe filein.dat fileout.res filenum.num 2
```

If the third parameter (language) is omitted, the English language will be used by default.

2.2 UNIX/MULTIVAC VERSION

Prior to program execution, it is necessary to create the input data file, as described in section §3.2.

To run interactively this version of the program (console application), you must have already done the assignment of the symbols list in Airbus DS Structural Analysis Office using the command:

```
. asignaciones_calculo
```

This instruction can be executed in the commands terminal prior to execute the program, but it is strongly recommended to include this statement in the files '.profile', and '.kshrc' (if working in Korn Shell) or '.bashrc' (if you work in Bash Shell). To do this, the following lines should be included in the file ".profile":

```
. asignaciones_calculo
ENV=$HOME/.kshrc
export ENV
BASH_ENV=$HOME/.bashrc
export BASH_ENV
```

In files '.kshrc' and '.bashrc' the following line shall be added:

```
. asignaciones_calculo
```

The instruction to run the program in interactive mode in the MULTIVAC server is:

```
$urco
```

The program asks for the version and language to use and the name of the input and output files, as shown in the following example:

```
Idioma / Language:
=====
1 ... ESPAÑOL
2 ... ENGLISH

Idioma / Language: 2

Name of the input file: Example.dat
```

```
Name of the output file: Example.res  
Name of the numeric file: Example.num  
Urco 5.4. (c) Airbus Defence & Space, 2017
```

The CPU time for calculation depends on the quantity of load cases but it is, in general, negligible.

In the same way as for the Windows console version, this application for Unix can be also executed in one single command, by passing as arguments the input data file (see sub-chapter §2.1.2):

```
$urco filein.dat fileout.res filenum.num 2
```

If the third parameter (language) is omitted, the English language will be used by default.

2.3 LINUX/MULTIVAC VERSION

To allow the use of this version, it is necessary to issue the same arrangements as for Unix version in the files '.profile', '.kshrc' and '.bashrc'. The Linux version is still run from the same Unix Solaris command line terminal in the Multivac3 server, even the Linux HPC cores will do the work in the background (a special wrapper acts as interface between these platforms).

The instruction to run the Linux version of the program in interactive mode in MULTIVAC server is:

```
urco
```

In the same way as for the Windows console and Unix versions, this console application for Linux can be also executed in one single command, by passing as arguments the input data file (see sub-chapters §2.1.2 and §2.2):

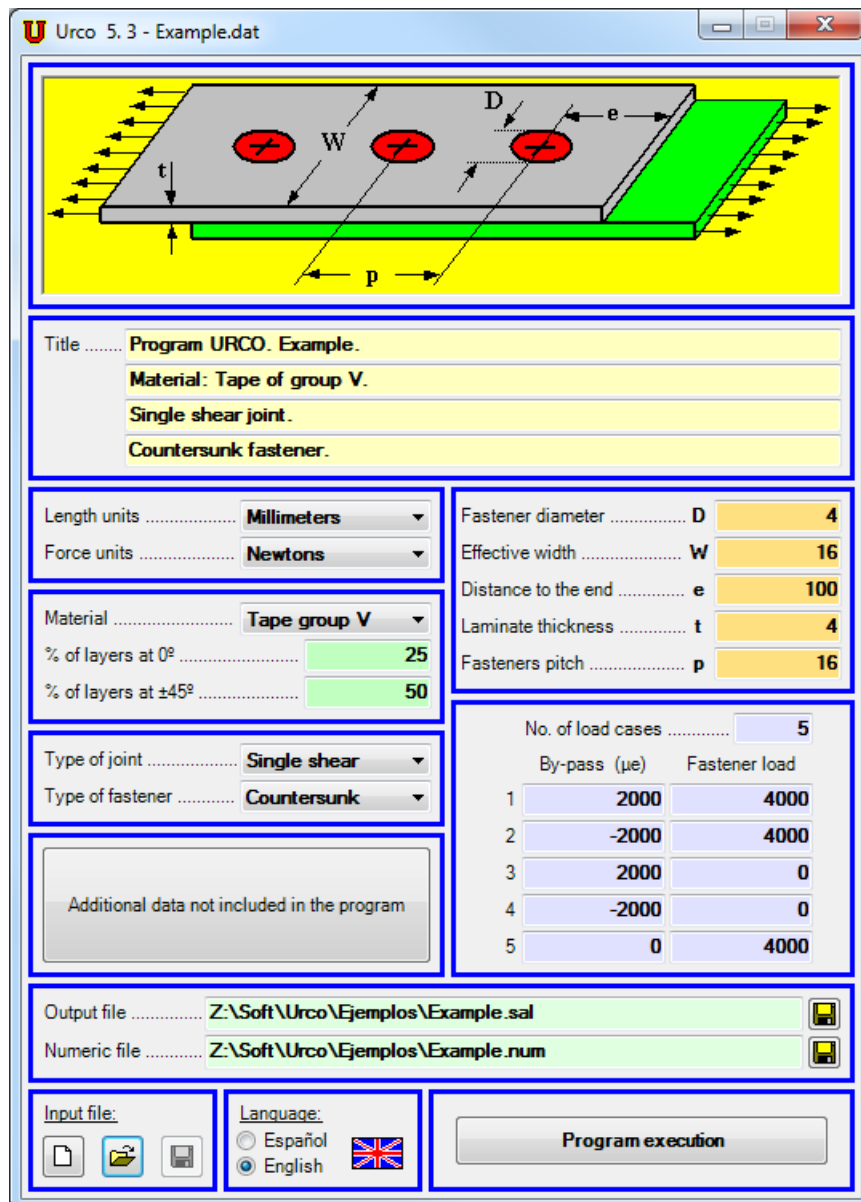
```
urco filein.dat fileout.res filenum.num 2
```

Notice that CPU time for calculation of Linux program version is expected to be lower than Unix one.

3 INPUT DATA

3.1 WINDOWS PC/GUI

The input data must be directly typed in the application form or retrieved from an input file after clicking  button in the main form, which has the following appearance, after loading the data corresponding to the example described in chapter 6:



URCO 5.3 - Example.dat

Diagram: A 3D perspective view of a rectangular laminate plate of thickness t and width W . It features three fasteners of diameter D spaced at a pitch p . The distance from the center of a fastener to the edge is e . The plate is shown with a green top surface and a yellow bottom surface.

Title: Program URCO. Example.

Material: Tape of group V.

Single shear joint.

Countersunk fastener.

Length units: Millimeters

Force units: Newtons

Material: Tape group V

% of layers at 0° : 25

% of layers at $\pm 45^\circ$: 50

Fastener diameter D : 4

Effective width W : 16

Distance to the end e : 100

Laminate thickness t : 4

Fasteners pitch p : 16

Type of joint: Single shear

Type of fastener: Countersunk

No. of load cases: 5

	By-pass (μe)	Fastener load
1	2000	4000
2	-2000	4000
3	2000	0
4	-2000	0
5	0	4000

Additional data not included in the program

Output file: Z:\Soft\Urco\Ejemplos\Example.sal

Numeric file: Z:\Soft\Urco\Ejemplos\Example.num

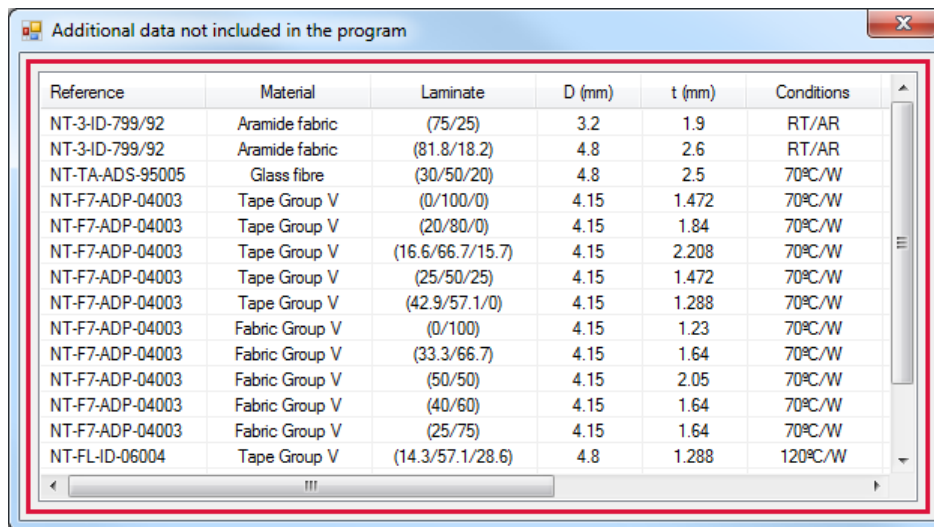
Input file:   

Language: ☐ Español ☒ English  

Program execution

Figure 1. URCO input data form.

Pressing the button labelled as 'Additional data not included in the program', a set of test results not included in the database of the program are displayed for information in the form shown in Figure 2.



Reference	Material	Laminate	D (mm)	t (mm)	Conditions
NT-3-ID-799/92	Aramide fabric	(75/25)	3.2	1.9	RT/AR
NT-3-ID-799/92	Aramide fabric	(81.8/18.2)	4.8	2.6	RT/AR
NT-TA-ADS-95005	Glass fibre	(30/50/20)	4.8	2.5	70°C/W
NT-F7-ADP-04003	Tape Group V	(0/100/0)	4.15	1.472	70°C/W
NT-F7-ADP-04003	Tape Group V	(20/80/0)	4.15	1.84	70°C/W
NT-F7-ADP-04003	Tape Group V	(16.6/66.7/15.7)	4.15	2.208	70°C/W
NT-F7-ADP-04003	Tape Group V	(25/50/25)	4.15	1.472	70°C/W
NT-F7-ADP-04003	Tape Group V	(42.9/57.1/0)	4.15	1.288	70°C/W
NT-F7-ADP-04003	Fabric Group V	(0/100)	4.15	1.23	70°C/W
NT-F7-ADP-04003	Fabric Group V	(33.3/66.7)	4.15	1.64	70°C/W
NT-F7-ADP-04003	Fabric Group V	(50/50)	4.15	2.05	70°C/W
NT-F7-ADP-04003	Fabric Group V	(40/60)	4.15	1.64	70°C/W
NT-F7-ADP-04003	Fabric Group V	(25/75)	4.15	1.64	70°C/W
NT-FL-ID-06004	Tape Group V	(14.3/57.1/28.6)	4.8	1.288	120°C/W

Figure 2. Additional data not included in the program.

For future analyses use, the introduced input data can be stored in a text file by clicking the  button located at the bottom left corner of the main form. In addition, all data introduced in this form can be cleared by clicking  button.

Notice that labels shown in the input data form have the same meanings as the ones defined in program input data text file. Detailed explanations of them can be found in the following sub-chapter.

3.2 INPUT FILE

3.2.1 INPUT DATA STRUCTURE

This file contains all the necessary data that define the problem, as well as the different options required to solving it.

Basically, it contains seven data groups, all of them are necessary. They will be numbered from 1 to 7 and must be in order.

All the numerical data contained in this file will be read in a free format. The variables will be separated by one or more blank spaces. The numbering of each group will be included in the first column. Listed below are the contents of each data group:

GROUP 1: Title of the problem:

The first four text lines may contain, for information purposes, the title of the problem to be solved:

- Line 1 → 1 First line of the title
- Line 2 → Second line of the title
- Line 3 → Third line of the title
- Line 4 → Fourth line of the title

GROUP 2: Units for length and forces:

This group has three text lines with the following information:

Line 1 → 2 'Optional commentary'

Line 2 → [Length]

Line 3 → [Force]

Where:

[Length] = This parameter contains a string to define the units for lengths. It must be one of the following:

mm = millimetres

cm = centimetres

in = inches

[Force] = It is another string to define the units for forces. It must be one of the following:

N = Newtons

kg = force kilograms

lb = force pounds

GROUP 3: Type of material:

The structure of this group is the following one:

Line 1 → 3 'Optional commentary'

Line 2 → *Nmat*

Where:

Nmat = Type of material. It is an integer value in the range from 1 to 6, and represents the type of the composite material to be considered by the program, according to these criteria:

1 = Fabric of Group III

2 = Tape of Group III

3 = Fabric of Group IV

4 = Tape of Group IV

5 = Fabric of Group V

6 = Tape of Group V

GROUP 4: Lay-up:

The data contained in this group depends on the type of material defined in group 3, fabric or tape, as shown below:

- **Fabric:**

Line 1 → 4 'Optional commentary'

Line 2 → $P_{0/90}$

- **Tape:**

Line 1 → 4 'Optional commentary'

Line 2 → P_0 P_{45}

Where:

$P_{0/90}$ = Percentage of fabric layers with 0°/90° orientation. It is assumed by the program that the rest of layers are +45°/-45° oriented.

P_0 = Percentage of tape layers with 0° orientation.

P_{45} = Total percentage of tape layers with +45° and -45° orientations. It is assumed by the program that the rest of layers are 90° oriented.

Remark: The program URCO version 5 only supports laminates made of layers of one composite material but, notice that it can be also obtained conservative URCO allowable values for laminates made of a mixture of two different layer materials (e.g. tape and fabric mixed laminates), according to the approach described in sub-chapter A.3.

GROUP 5: Type of joint and fastener:

This group contains two text lines with the following information:

Line 1 → 5 'Optional commentary'

Line 2 → N_j N_f

Where:

N_j = Type of joint. It is an integer value with the following meanings:

1 = Single shear

2 = Double shear

N_f = Type of fastener. It is an integer value that can be one of the following:

1 = Protruding fastener

2 = Countersunk fastener

The option $Nj = 1$ must be selected when the element to be analyzed is in a single shear joint or it is an external plate in a double shear joint:

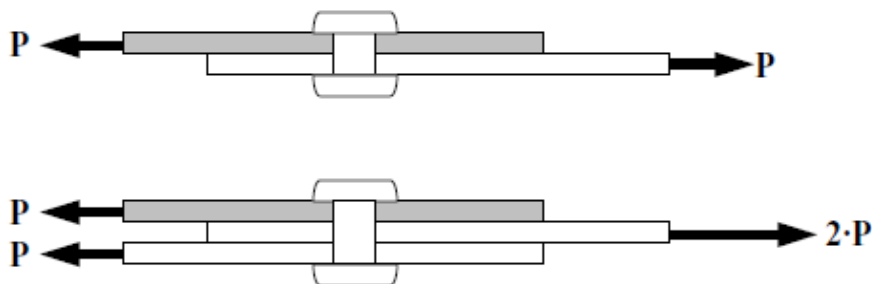


Figure 3. Elements highlighted in grey colour are in single shear.

The option $Nj = 2$ must be selected when the element to be analyzed is the central one in a double shear joint:



Figure 4. Element highlighted in grey colour is in double shear.

The option $Nf = 2$ (countersunk fastener) applies only if the analyzed element is in single shear (single shear joint or external element in a double shear joint) and the fastener has the countersunk head in the considered element. Otherwise $Nf = 1$ must be specified.

GROUP 6: Geometrical Data:

This group has two text lines with the following data:

Line 1 → 6 'Optional commentary'

Line 2 → $D \quad W \quad e \quad t \quad p$

Where:

D = Fastener diameter.

W = Effective width of the laminate.

e = Distance to the edge of the joint in the load direction.

t = Laminate thickness.

p = Fastener pitch in the load direction.

GROUP 7: Load cases:

The arrangement of this group is the following one:

Line 1	→	7	'Optional commentary'	
Line 2	→	N_c	$\varepsilon(1)$	$P(1)$
Line 3	→		$\varepsilon(2)$	$P(2)$
				
				
Line $N_c + 1$	→		$\varepsilon(N_c)$	$P(N_c)$

Where;

N_c = Number of load cases for which the reserve factors are going to be calculated.

$\varepsilon(i)$ = By-pass load in terms of micro strains in load case ' i '.

$P(i)$ = Shear load transferred through the fastener in load case ' i '.

If $N_c = 0$, only the joint allowable values **A**, **B** and **C** will be calculated by the program.

3.2.2 GEOMETRICAL DATA GENERAL CRITERIA

The general criteria for the determination of several of the above parameters (effective width, distance to the edge and fasteners pitch) are explained below:

Reference joint configuration:

The configuration of the test specimens used to obtain the allowable values was the following one:

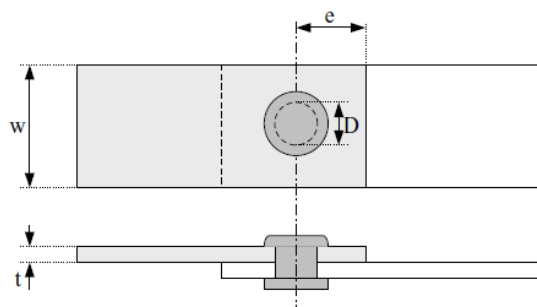


Figure 5. Reference joint configuration scheme.

In this case, for the analysis of the element shown at left side in Figure 5, it must be defined:

- Effective width of the laminatew
- Distance to the edge of the joint in the load direction.....e
- Fastener pitch in the load direction.....N/A

Tensile lap shear general joint configuration:

A more general type of tensile lap shear joint than previous one is shown in the following figure:

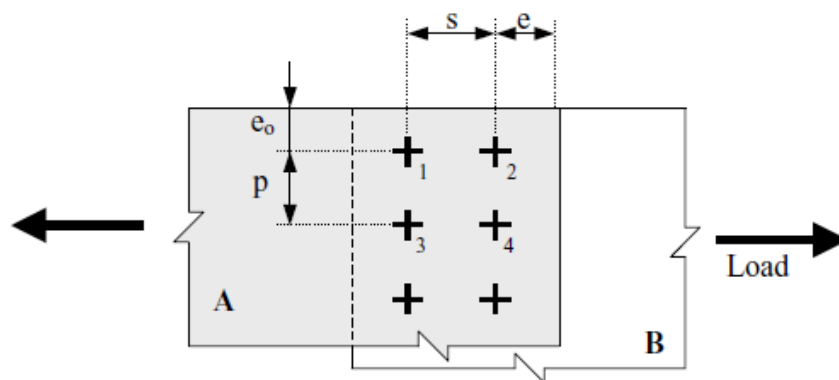


Figure 6. Tensile lap shear general joint configuration scheme.

In this case, for the analysis of the element (A) shown at left side in Figure 6, the input geometrical parameters for each fastener must be defined according to the following table:

Fastener	URCO input data		
	Effective width	Edge distance	Fastener pitch
#1	$\min(p, 2 \cdot e_0)$	N/A	s
#2	$\min(p, 2 \cdot e_0)$	e	N/A
#3	p	N/A	s
#4	p	e	N/A

Table 1. Tensile lap shear general joint configuration URCO input data.

Tensile lap shear staggered pitch joint configuration:

A scheme of this type of joint can be seen in more detail in the following figure:

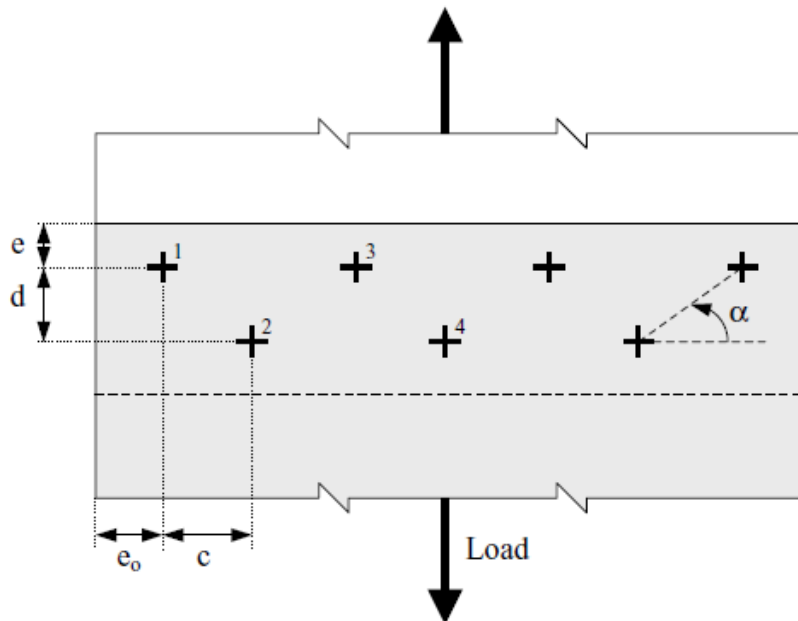


Figure 7. Tensile lap shear staggered pitch joint configuration scheme.

In this case, for the analysis of the element shown at bottom in Figure 7, the input geometrical parameters for each fastener must be defined according to the following table:

Fastener	URCO input data		
	Effective width	Edge distance	Fastener pitch
#1	$\min(c, 2 \cdot e_0)$	e	N/A
#2	c	N/A	d
#3	c	e	N/A
#4	c	N/A	d

Table 2. Tensile lap shear staggered pitch joint configuration URCO input data.

Remark: This criterion is too conservative for staggered patterns with $\alpha > 60^\circ$. In such case, the joint configuration can be considered as two parallel rows of fasteners.

Shear load staggered pitch joint configuration:

A scheme of this type of joint can be seen in more detail in the following figure:

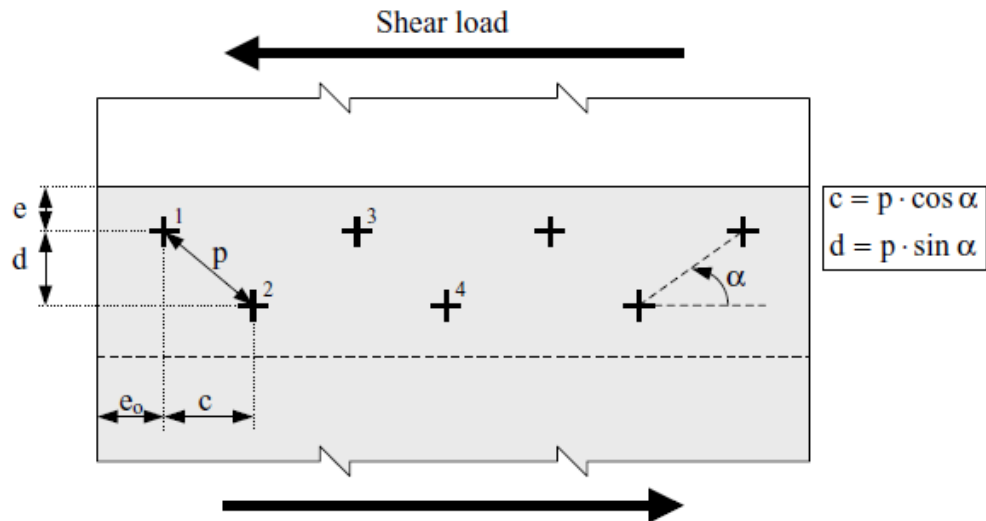


Figure 8. Shear load staggered pitch joint configuration scheme.

In this case, for the analysis of the element shown at bottom in Figure 7, the input geometrical parameters for each fastener must be defined according to the following table:

Fastener	URCO input data			
	Effective width		Edge distance	Fastener pitch
	$\alpha < 30^\circ$	$\alpha \geq 30^\circ$		
#1	$2 \cdot e$	$\min(p, 2 \cdot e)$	e_0	N/A
#2	$2 \cdot (d+e)$	p	N/A	c
#3	$2 \cdot e$	$\min(p, 2 \cdot e)$	N/A	c
#4	$2 \cdot (d+e)$	p	N/A	c

Table 3. Shear load staggered pitch joint configuration URCO input data.

Remark: The results obtained in this case can be conservative or not due to the following reasons:

- For the same fastener load, the stress concentration due to the shear transfer is, in general, higher than for the axial load transfer (no conservative).
- The point of higher stress due to bearing is different from the corresponding to the by-pass load (conservative).

Geometrical input data general remarks:

In all above mentioned configurations, when a parameter does not apply (N/A) or it is infinite (value enough big from a practical point of view), a null value (0) must be introduced in the program input data file.

Notice that the present chapter explains which effective width values should be considered in standard joints analyses but all of them correspond to design configurations with rivets of same diameter, homogeneous pitch between rivets and well known applied loads directions; therefore, in order to support the URCO program input data definition by the user, a methodology of how to calculate the joint effective width at fastener in more generic design configurations has been included in APPENDIX A: URCO ADDITIONAL APPLICATIONS.

4 OUTPUT RESULTS

4.1 WINDOWS/PC GUI

Pressing the button 'Program Execution' at the input data application form (see Figure 1), the program performs the analysis, writes the output files and provides the output results by means of the following form, which corresponds to the input data of the example described in chapter 6:

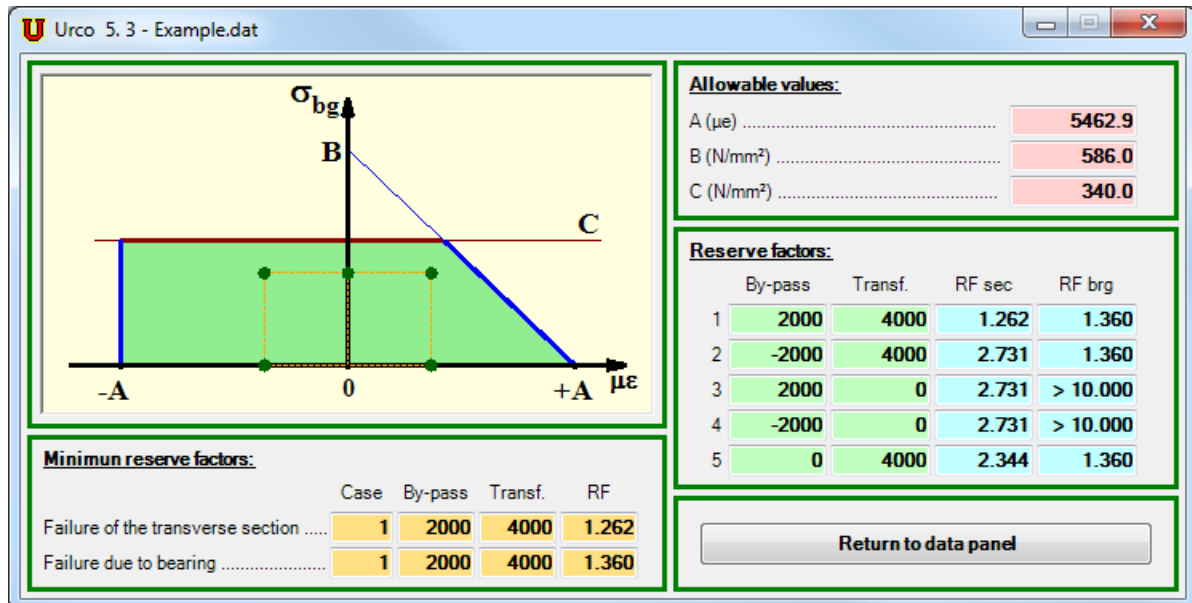


Figure 9. URCO output results form.

The diagram located in the left upper corner of the results form shows the safe region (marked in green colour) for every bearing stress and by-pass strain combination, attending to the allowable values **A**, **B** and **C** obtained by the program. The diagram also plots several black points which represent the analysed load cases.

Labels shown in output forms have the same meanings as in the output text files. Detailed explanations can be found in following sub-chapter.

4.2 OUTPUT FILES

This sub-chapter includes the description of the two output data text files provided by URCO program in all the platforms (Windows/PC GUI, Windows/PC console and UNIX/LINUX MULTIVAC consoles):

- Results file.
- Numeric file.

See following paragraphs for more details.

4.2.1 RESULTS FILE

The output file includes the following information:

Summary of the input data:

- Title.
- Material type lay-up.
- Joint and fastener type.
- Geometry of the joint.

Allowable values:

- **A:** Maximum allowed by-pass load, expressed in terms of microstrains, for tension or compression failure of the laminate at the gross cross section located behind the fastener hole.
- **B:** Maximum allowed bearing stress for failure of the laminate due to tension in the net cross section located at the fastener hole.
- **C:** Maximum allowed bearing stress for failure of the joint due to bearing of the laminate at the fastener hole.

Reserve factors:

- Reserve factors in all load cases, for the following failure modes:
 - Tension or Compression strength of the laminate at the net cross section of the fastener hole (By-pass/Tension-Through-the-Hole interaction strength).
 - Bearing strength at fastener hole.
- Minimum reserve factors, indicating their corresponding load case, for both considered failure modes.

4.2.2 NUMERIC FILE

This file contains only the results obtained from analysis, without any label. The information contained is the following one:

- Allowable values: Points *A*, *B* and *C*.
- Number of load cases: N_c
- Load cases with their corresponding reserve factors: i , $\varepsilon(i)$, $P(i)$, $RF_1(i)$, $RF_2(i)$
- Load case and minimum reserve factor of failure in the net cross section: N_1min , RF_1min
- Load case and minimum reserve factor of failure due to bearing: N_2min , RF_2min

This file is created to be easily used as input for other programs and utilities. Its arrangement is the following one:

A	B	C		
Nc				
1	$\varepsilon(1)$	$P(1)$	$RF_1(1)$	$RF_2(1)$
2	$\varepsilon(2)$	$P(2)$	$RF_1(2)$	$RF_2(2)$
...				
...				
Nc	$\varepsilon(Nc)$	$P(Nc)$	$RF_1(Nc)$	$RF_2(Nc)$

4.3 WARNING AND ERROR MESSAGES

In console versions, when the program detects an error in the input data or writing the results, it writes the corresponding message in the console and aborts the execution of the program.

If some value in the data exceeds the range of available test results considered for the development and validation of the program, the corresponding warning is displayed on the screen and the program continues the execution.

The error messages that might appear in the console are listed below:

English output error messages:

- ERROR #1: The input file cannot be opened.
- ERROR #2: Erroneous group number in the input file.
- ERROR #3: Group 1 missing in input file (Title).
- ERROR #4: End of file while reading group 1 in input file (Title).
- ERROR #5: Group 2 missing in input file (Units).
- ERROR #6: End of File or error while reading group 2 in input file (Units).
- ERROR #7: Units no supported by the program.
- ERROR #8: Group 3 missing in input file (Material).
- ERROR #9: End of file or error while reading group 3 in input file (Material).

-
- ERROR #10: Selection material out of bounds.
- ERROR #11: Group 4 missing in input file (Layers).
- ERROR #12: End of file or error reading group 4 (Layers).
- ERROR #13: Percentage of layers at 0° for tape must be in the range of 0 to 100.
- ERROR #14: Percentage of layers at $0^\circ/90^\circ$ for fabric must be in the range of 0 to 100.
- ERROR #15: Percentage of layers at 45° must be in the range of 0 to 100.
- ERROR #16: Percentage of layers at $+45^\circ/-45^\circ$ must be in the range of 0 to 100.
- ERROR #17: Percentage (0°) + Percentage (45°) for tape must be lower than 100%.
- ERROR #18: Group 5 missing in input file (Joint).
- ERROR #19: End of file or error while reading group 5 (Joint).
- ERROR #20: Selection out of bounds single/double shear.
- ERROR #21: Selection out of bounds protruding/countersunk.
- ERROR #22: Group 6 missing in input file (Geometry).
- ERROR #23: End of file or error while reading group 6 (Geometry).
- ERROR #24: The diameter must be positive.
- ERROR #25: The width must be positive.
- ERROR #26: Distance to the end of the joint must be positive.
- ERROR #27: The laminate thickness must be positive.
- ERROR #28: Fastener pitch must be positive.
- ERROR #29: The width must be bigger than the diameter.
- ERROR #30: The distance to the end must be bigger than the radius.
- ERROR #31: The fastener pitch must be bigger than the diameter.
- ERROR #32: Group 7 missing in input file (Loads).
- ERROR #33: End of file or error while reading group 7 in input file (Loads).
- ERROR #34: The number of load cases exceeds the maximum of the program.
-

ERROR #35: Transferred load through the fastener must be positive.

ERROR #36: The output file cannot be opened.

ERROR #37: Writing error in the output file.

ERROR #38: The numeric output file cannot be opened.

ERROR #39: Writing error in the numeric output file.

Spanish output error messages:

ERROR #1: No se puede abrir el fichero de entrada.

ERROR #2: Número de grupo erróneo en el fichero de entrada.

ERROR #3: Grupo 1 ausente en el fichero de entrada (Título).

ERROR #4: Fin de fichero mientras se leía el grupo 1 en el fichero de entrada (Título).

ERROR #5: Grupo 2 ausente en el fichero de entrada (Unidades).

ERROR #6: Fin de fichero o error mientras se leía el grupo 2 (Unidades).

ERROR #7: Unidades no soportadas por el programa.

ERROR #8: Grupo 3 ausente en el fichero de entrada (Material).

ERROR #9: Fin de fichero o error mientras se leía el grupo 3 (Material).

ERROR #10: Selección material fuera de límites.

ERROR #11: Grupo 4 ausente en el fichero de entrada (Apilado).

ERROR #12: Fin de fichero o error mientras se leía el grupo 4 (Apilados).

ERROR #13: Porcentaje de capas a 0° para cinta debe estar comprendido entre 0 y 100.

ERROR #14: Porcentaje de capas a 0°/90° para tejido debe estar comprendido entre 0 y 100.

ERROR #15: Porcentaje de capas a 45° debe estar comprendido entre 0 y 100.

ERROR #16: Porcentaje de capas a +45°/-45° debe estar comprendido entre 0 y 100.

ERROR #17: Porcentaje (0°) + Porcentaje (45°) para cinta debe ser menor que 100%.

-
- ERROR #18: Grupo 5 ausente en el fichero de entrada (Unión).
- ERROR #19: Fin de fichero o error mientras se leía el grupo 5 (Unión).
- ERROR #20: Selección fuera límites simple/doble cortadura.
- ERROR #21: Selección fuera límites protuberante/avellanado.
- ERROR #22: Grupo 6 ausente en el fichero de entrada (Geometría).
- ERROR #23: Fin de fichero o error mientras se leía el grupo 6 (Geometría).
- ERROR #24: El diámetro tiene que ser positivo.
- ERROR #25: La anchura debe ser positiva.
- ERROR #26: La distancia al extremo de la unión debe ser positiva.
- ERROR #27: El espesor del laminado debe ser positivo.
- ERROR #28: El paso entre remaches debe ser positivo.
- ERROR #29: La anchura debe ser mayor que el diámetro.
- ERROR #30: La distancia al extremo debe ser mayor que el radio.
- ERROR #31: El paso entre remaches debe ser mayor que el diámetro.
- ERROR #32: Grupo 7 ausente en el fichero de entrada (Cargas).
- ERROR #33: Fin de fichero o error mientras se leía el grupo 7 (Cargas).
- ERROR #34: El número de casos de carga excede el máximo del programa.
- ERROR #35: La carga transferida por el remache debe ser positiva.
- ERROR #36: No se puede abrir el fichero de salida.
- ERROR #37: Error de escritura en el fichero de salida.
- ERROR #38: No se puede abrir el fichero numérico de salida.
- ERROR #39: Error de escritura en el fichero numérico de salida.

English output warning messages:

- WARNING #1: Available data are only for percentage of layers at 0° between 10% and 60%. Points A and B will be calculated by extrapolation.

-
- WARNING #2: Available data are only for percentage of layers at $0^{\circ}/90^{\circ}$ between 20% and 100%. Points A and B will be calculated by extrapolation.
- WARNING #3: Available data are only for diameters between 4 and 12.7 millimeters. Points A and B will be calculated by extrapolation. The results could be UNCONSERVATIVE.
- WARNING #4: Available data are only for effective widths greater than 2 times the diameter. Point B will be calculated by extrapolation.
- WARNING #5: Available data are only for distances to the end greater than 1.8 times the diameter. Point B will be calculated by extrapolation.
- WARNING #6: Available data are only for thicknesses between 0.7 and 1.5 times the diameter. Point B will be calculated by extrapolation.
- WARNING #7: Available data are only for thicknesses between 0.5 and 1.5 times the diameter. Point B will be calculated by extrapolation.
- WARNING #8: Available data are only for fastener pitches greater than 1.8 times the diameter. Point B will be calculated by extrapolation.
- WARNING #9: The Pitch/Diameter ratio does not match with the design principles. It is recommended: Pitch/Diameter > 4

Spanish output warning messages:

- AVISO #1: Sólo se dispone de datos para porcentajes de capas a 0° comprendidos entre el 10% y el 60%. Los puntos A y B serán calculados por extrapolación.
- AVISO #2: Sólo se dispone de datos para porcentajes de capas a $0^{\circ}/90^{\circ}$ comprendidos entre el 20% y el 100%. Los puntos A y B serán calculados por extrapolación.
- AVISO #3: Sólo se dispone de datos para diámetros comprendidos entre 4 y 12.7 milímetros. Los puntos A y B serán calculados por extrapolación. Los resultados podrían ser NO CONSERVATIVOS.
- AVISO #4: Sólo se dispone de datos para anchuras efectivas mayores que 2 veces el diámetro. El punto B será calculado por extrapolación.
- AVISO #5: Sólo se dispone de datos para distancias al extremo mayores que 1.8 veces el diámetro. El punto B será calculado por extrapolación.

- AVISO #6: Sólo se dispone de datos para espesores comprendidos entre 0.7 y 1.5 veces el diámetro. El punto B será calculado por extrapolación.
- AVISO #7: Sólo se dispone de datos para espesores comprendidos entre 0.5 y 1.5 veces el diámetro. El punto B será calculado por extrapolación.
- AVISO #8: Sólo se dispone de datos para pasos entre remaches mayores que 1.8 veces el diámetro. El punto B será calculado por extrapolación.
- AVISO #9: La relación Paso/Diámetro no cumple las condiciones de diseño. Se recomienda: Paso/Diámetro > 4

5 LIMITATIONS

The methodology followed by the program is based in coupon tests; therefore, its applicability should be restricted to the range of design configurations covered by these tests. Within certain limits, it also proposes allowable values beyond the design space strictly covered by tests. The user is informed in such cases with warning messages, as listed in sub-chapter 4.3.

Therefore, taking all herein above into account, the results provided by the program execution can be directly used to analyze joint design configurations that fulfil with the following conditions:

- 1) Maximum temperature of 70°C and maximum humidity of 80%RH at the analysed joint.
- 2) Maximum quantity of applied load cases: 255.
- 3) Composite materials of the joined elements:
 - Fabric Group III (CASA designation: Z-19.749)
 - Tape Group III (CASA designation: Z-19.766)
 - Fabric Group IV (CASA designation: Z-19.776)
 - Tape Group IV (CASA designation: Z-19.785)
 - Fabric Group V (CASA designation: Z-19.760)
 - Tape Group V (CASA designation: Z-19.775)
- 4) Fastener diameter shall be between 4.0 and 12.7 millimetres.
- 5) Fastener type can be protruding or countersunk.
- 6) Percentage of 0° layers in tape laminates shall be between 10% and 60%, included.
- 7) Percentage of 0°/90° layers in fabric laminates shall be between 20% and 100%, included.
- 8) Ratio between effective width and fastener diameter shall be: $w/D \geq 2.0$
- 9) Ratio between edge distance and fastener diameter shall be: $e/D \geq 1.8$
- 10) Ratio between fasteners pitch and fastener diameter shall be: $p/D \geq 1.8$
- 11) In single shear analyses, the ratio between laminate thickness and fastener diameter (t/D) shall be inside the following ranges:
 - Protruding fasteners: $0.7 \leq t/D \leq 1.5$
 - Countersunk fasteners: $0.5 \leq t/D \leq 1.5$

6 EXAMPLE

The case to be solved consists in an axial loaded joint between two plates, fastened each other, as shown in the following figure:

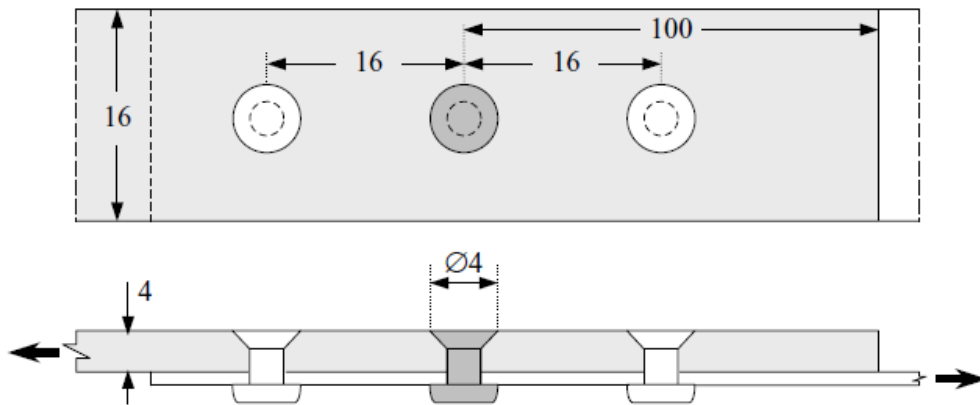


Figure 10. URCO example geometry data.

The element to be studied (the one highlighted in grey colour in Figure 10) is supposed to be made in 8552/AS4 tape (Group V material), with a typical orthotropic lay-up sequence (25/50/25).

The fastener hole under study will be the central one. The load transferred through the fastener as well as the by-pass strain at the hole must be determined by means of a load transfer analysis (i.e. by running an ARES program analysis).

In this example the following five load cases will be taken into account:

Load case	By-pass strain ($\mu\epsilon$)	Transferred load by fastener (N)
#1	2000	4000
#2	-2000	4000
#3	2000	0
#4	-2000	0
#5	0	4000

Table 4. URCO example applied loads.

With all the above data, therefore it is possible to generate the input data file to run the analysis. This file as well as the obtained results and numeric output text files are provided on next paragraphs.

Example input file:

```

1 Program URCO. Example.
  Material: Tape of group V.
  Single shear joint.
  Countersunk fastener.
2 UNITS: [Length] / [Force]
  mm
  N
3 TYPE OF MATERIAL: Fabric: 1,3,5 - Tape: 2,4,6
  6
4 LAY-UP: Tejido: %0/90 - Cinta: %0 y %45
  25.0 50.0
5 TYPE OF JOINT AND FASTENER: Single=1 Double=2 - Protruding=1 Countersunk=2
  1 2
6 GEOMETRY: Diameter - Width - Distance end - Thickness - Pitch
  4.0 16.0 100.0 4.0 16.0
7 LOAD CASES: Ncases, By-pass(i), Transferred(i)
  5      2000.0      4000.0
      -2000.0      4000.0
      2000.0        0.0
      -2000.0        0.0
      0.0           4000.0
  
```

Example output results file:

```

*****
*                                     *
*      <<< U R C O >>>              *
*                                     *
*****
* Calculation of allowable values *
* in composite fastened joints   *
*****

<--+-----+-----+-----+-----+
<--|          |<-- p -->|<-- e -->|   |
<--|          |          P          |   |
<--|    0      0          0==>      |   W
<--|          |          |          |   |
<--|          |          |          |   |
<--+-----+-----+-----+-----+

Title:
=====

Program URCO. Example.
Material: Tape of group V.
Single shear joint.
Countersunk fastener.
  
```


Material data:

=====

Material: Tape Group V

% of layers of fibres at 0° ...: 25.00

% of layers of fibres at 45° ...: 50.00

% of layers of fibres at 90° ...: 25.00

Joint data:

=====

Type of joint: Single Shear

Type of fastener: Countersunk

Geometry of the joint:

=====

Fastener diameter: D = 4.00 mm

Effective width: W = 16.00 mm

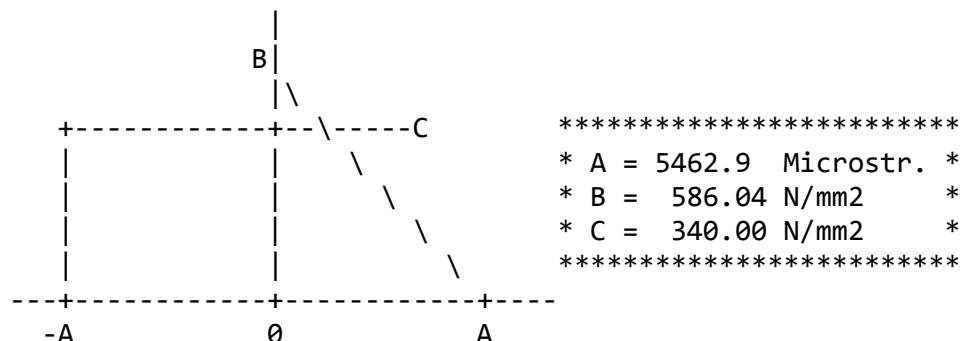
Distance to the end: e = 100.00 mm

Laminate thickness: t = 4.00 mm

Fasteners pitch: p = 16.00 mm

Allowable values:

=====



Reserve factors:

=====

Case	Load		Reserve factor	
	By-pass (μe)	Transf. (N)	Section	Bearing
1	2000.0	4000.0	1.26	1.36
2	-2000.0	4000.0	2.73	1.36
3	2000.0	0.0	2.73	> 10.00
4	-2000.0	0.0	2.73	> 10.00
5	0.0	4000.0	2.34	1.36

Minimum reserve factors:

=====

Cross section failure: Rf = 1.26
 Load Case: 1
 By-pass microstrains: e = 2000.0
 Transferred load: P = 4000.0 N

Failure due to bearing: Rf = 1.36
 Load Case: 1
 By-pass microstrains: e = 2000.0
 Transferred load: P = 4000.0 N

Program Urco. Version 5.4. (c) Airbus Defence & Space, 2017

Example output numeric file:

5462.878	586.0367	340.0000		
5				
1	2000.000	4000.000	1.261508	1.360000
2	-2000.000	4000.000	2.731439	1.360000
3	2000.000	0.000000E+00	2.731439	1.000000E+32
4	-2000.000	0.000000E+00	2.731439	1.000000E+32
5	0.000000E+00	4000.000	2.344147	1.360000
1	1.261508			
1	1.360000			

APPENDIX A: URCO ADDITIONAL APPLICATIONS

A.1 INTRODUCTION

The present appendix includes several analysis guidelines for additional applications of URCO program output data in terms of solving typical design configurations outside the standard use of the software.

Notice that these guidelines describe methodologies that are not directly implemented in URCO therefore, they shall be considered as an independent post-processing to be performed by the user of the output results provided by the program so, their applicability or validity may be conditioned by any other design requirements outside URCO software environment. According to this, the following guidelines must be applied with caution.

A.2 REFERENCES

	<u>Document</u>	<u>Issue</u>	<u>Title</u>
Ref. 5	NT-T-ADS-92042	B	'Procedimiento de Cálculo de Uniones Remachadas Metálicas' Technical Report. [Original in Spanish]
Ref. 6	[Airbus] V020RP0818061	6.0	'Static Method for Sizing of Composite Bolted Joints – Exportable Version' Technical Report.

A.3 ANALYSES WITH MIXED LAMINATES

The program URCO version 5 only supports laminates made of layers of same composite material. However, it can be also obtained conservative URCO allowable values for laminates made of a mixture of two different layer materials, according to the approach described in this sub-chapter.

In those configurations, two URCO complementary analyses shall be performed, one per each different base material involved in the laminate stacking sequence (e.g. Fabric G.IV and Tape G.IV), in order to obtain a conservative envelope of allowable values. Then, these two analyses shall consider the same input data (thickness, fastener diameter, etc.), except the laminate base material.

The percentages of fibre-layers per load main directions must be calculated according to the following expressions:

Tape introduced as URCO base material:

Allowable values A_{tape} , B_{tape} and C_{tape} will be obtained according to the following fibre-layers percentages:

$$0^\circ \text{ layers } \% = \frac{\sum \left[(n_{tape} \cdot t_{tape})_{0^\circ} + \frac{1}{2} (n_{fabric} \cdot t_{fabric})_{0^\circ/90^\circ} \right]}{t_{total}}$$

$$\pm 45^\circ \text{ layers \%} = \frac{\sum \left[(n_{\text{tape}} \cdot t_{\text{tape}})_{\pm 45^\circ} + (n_{\text{fabric}} \cdot t_{\text{fabric}})_{\pm 45^\circ} \right]}{t_{\text{total}}}$$

Fabric introduced as URCO base material:

Allowable values A_{fabric} , B_{fabric} and C_{fabric} will be obtained according to the following fibre-layers percentages:

$$\begin{aligned} 0^\circ/90^\circ \text{ layers \%} &= \min \left\{ \frac{\sum \left[2 \cdot (n_{\text{tape}} \cdot t_{\text{tape}})_{0^\circ} + (n_{\text{fabric}} \cdot t_{\text{fabric}})_{0^\circ/90^\circ} \right]}{t_{\text{total}}} ; 100\% \right\} \\ &= \min (2 \cdot 0^\circ \text{ layers \%} ; 100\%) \\ \pm 45^\circ \text{ layers \%} &= \frac{\sum \left[(n_{\text{tape}} \cdot t_{\text{tape}})_{\pm 45^\circ} + (n_{\text{fabric}} \cdot t_{\text{fabric}})_{\pm 45^\circ} \right]}{t_{\text{total}}} \end{aligned}$$

Where n_{tape} is the quantity of laminate tape layers in the indicated load direction (0° or $\pm 45^\circ$), n_{fabric} is the quantity of laminate fabric layers in the indicated load direction ($0^\circ/90^\circ$ or $\pm 45^\circ$), t_{tape} is the thickness of one tape layer, t_{fabric} is the thickness of one fabric layer and t_{total} is the total thickness of the laminate.

Once the allowable values A , B and C are obtained with URCO for each one of the base materials, the minimum ones can be taken as conservative allowable values for the analysed mixed laminate, as shown in the following expressions:

$$\text{Mixed laminate allowable} \rightarrow A = \min(A_{\text{tape}}, A_{\text{fabric}})$$

$$\text{Mixed laminate allowable} \rightarrow B = \min(B_{\text{tape}}, B_{\text{fabric}})$$

$$\text{Mixed laminate allowable} \rightarrow C = \min(C_{\text{tape}}, C_{\text{fabric}})$$

Remarks and limitations:

This methodology is only applicable if the percentage of fibre-layers in $\pm 45^\circ$ load direction of total laminate is $\leq 75.0\%$ for Group III or Group IV tape/fabric mixed laminates. In case of Group V tape/fabric mixed laminates, the percentage of fibre-layers in $\pm 45^\circ$ load direction of total laminate shall be $\leq 70.0\%$. If those limits are exceeded, the C allowable value obtained following this methodology may not be considered valid for mixed laminates analyses; for more details, contact with Stress Methods department.

Notice that user does not need to introduce as input data the $\pm 45^\circ$ fibre-layers percentage for fabric laminates analyses performed with URCO (v.5.2 and later) since this value is automatically calculated by the program using the $0^\circ/90^\circ$ fibre-layers percentage value previously introduced by the user. Therefore, it is important to remark that the $\pm 45^\circ$ fibre-layers percentage value automatically calculated by the program in fabric-based analyses (and shown e.g. in the output data file or GUI interface) may not match with the expected value (the one that can be calculated with above mentioned equations) when analysing fabric/tape mixed laminates, since in those cases, the quantity of fibre-layers at 0° may not be equal to the ones at 90° direction, as occur in conventional fabric

laminates. Therefore, if the conditions mentioned in previous paragraph are not exceeded ($\% \pm 45^\circ \leq 75.0$ or ≤ 70.0), this mismatch can be ignored by the user and the reported URCO allowable values could be considered valid to obtain the mixed laminate final ones.

A.4 IMPACT OF NON STRUCTURAL SHIMMING

The standard methods and allowable values available for the strength check of riveted joints can be applied to joints with non-structural shims with a maximum thickness (t_{shim}) up to the following limits:

- Liquid shim up to 0.5 mm thickness.
- Solid shim up to 1.2 mm thickness.

That approach is supported by past experience and tests and, as for fatigue issues mitigation in metallic plates, it is also recommended that shim total thickness does not exceed 25% of fastener diameter, see Ref. 5.

In case the non-structural shimming exceeds the above mentioned limitations and in absence of an internal dedicated test data, it is noted that Airbus Commercial proposes knock-down factors to apply to nominal design values in Ref. 6 (*Technical Report V020RP0818061 issue 6.0 'Static Method for Sizing of Composite Bolted Joints'*) for shim thicknesses up to 3.0 mm.

A.5 CALCULATION OF GENERIC EFFECTIVE WIDTH

Sub-chapter 3.2.2 explains which effective width should be considered in standard joints analyses but all of them correspond to design configurations with rivets of same diameter, homogeneous pitch between rivets and well known applied loads directions; therefore, in order to support the URCO program input data definition by the user, the following paragraphs describe a methodology, conservatively based in sub-chapter 12.1 of Ref. 6, of how to calculate the joint effective width at fastener in more generic design configurations than the ones covered by sub-chapter 3.2.2. In a first overview, this approach may be considered too conservative since the obtained value of effective width at the analysed fastener is valid for any applied load direction, but it should be remarked that this method can be also used to calculate the effective width of a joint by balancing the different diameters of the holes so, it can be very useful in non-standard designs or bi-axially loaded analyses.

Then, according to a linear weighting of fasteners diameter influence, the effective width w at any fastener of a generic riveted joint can be calculated using the following expression:

$$w = w_i = 2 \cdot \min \left(e_1^i ; e_2^i ; e_3^i ; e_4^i ; p_{ij} \cdot \frac{D_i}{D_i + D_j} \right)$$

Where:

w_i	Effective width of joint at Fastener i [mm]
$e_1^i , e_2^i , e_3^i , e_4^i$	Edge distances of Fastener i as indicated in Figure 11 for Fastener A [mm]
p_{ij}	Pitch between Fasteners i and j as indicated in Figure 11 for Fastener A [mm]

D_i, D_j Diameter of holes at Fasteners i and j , respectively [mm]

Figure 11 shows a generic riveted joint scheme with the area of influence of each fastener (blue circles), defined by the corresponding effective width calculated following this method, w_A , w_B and w_C .

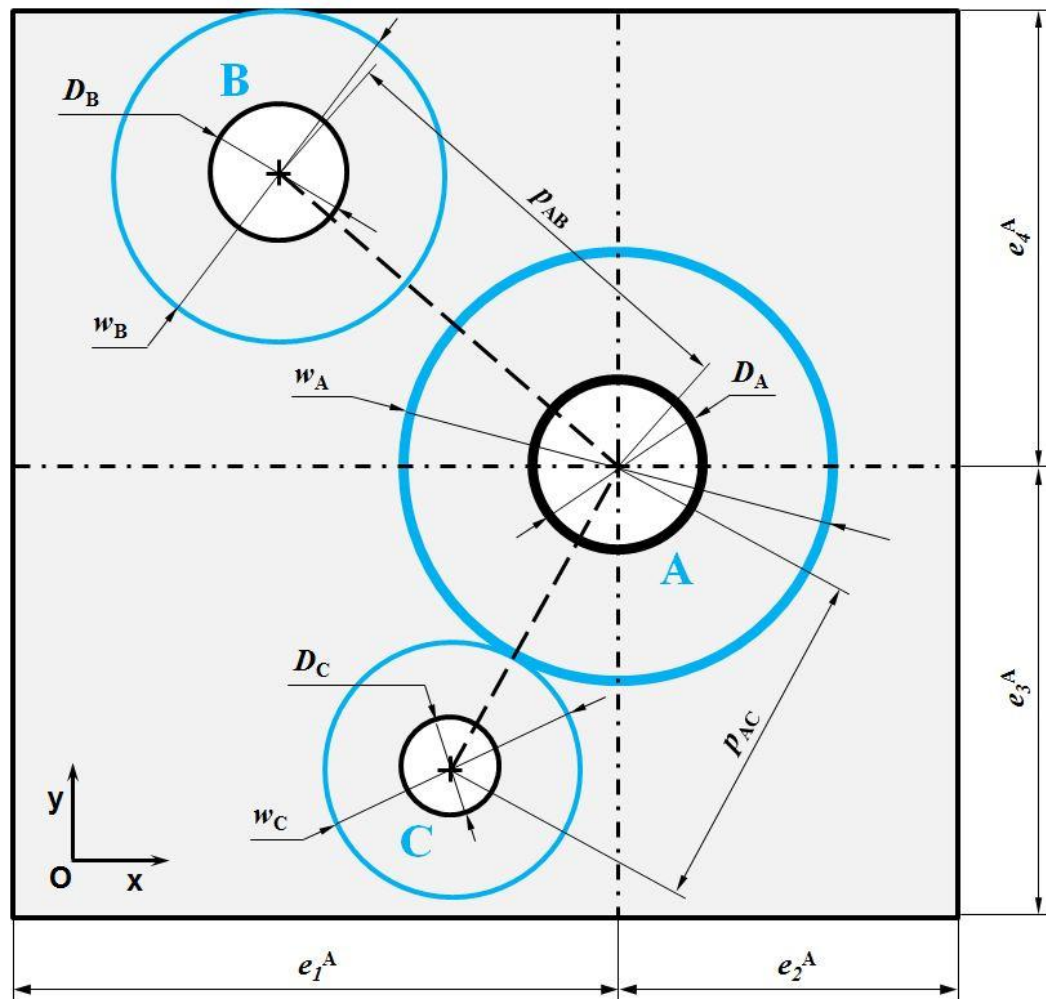


Figure 11. Geometrical parameters for Fastener A effective width calculation.

A.6 ANALYSES WITH REPAIR BUSHINGS

Allowable values calculation approach:

The methodology described in this sub-chapter can be used to obtain, by means of several complementary URCO analyses, the joint composite allowable values for joints where a bushing has been installed for repair purposes at the analyzed location. Figure 12 shows the different basic types of bushings covered by this methodology.

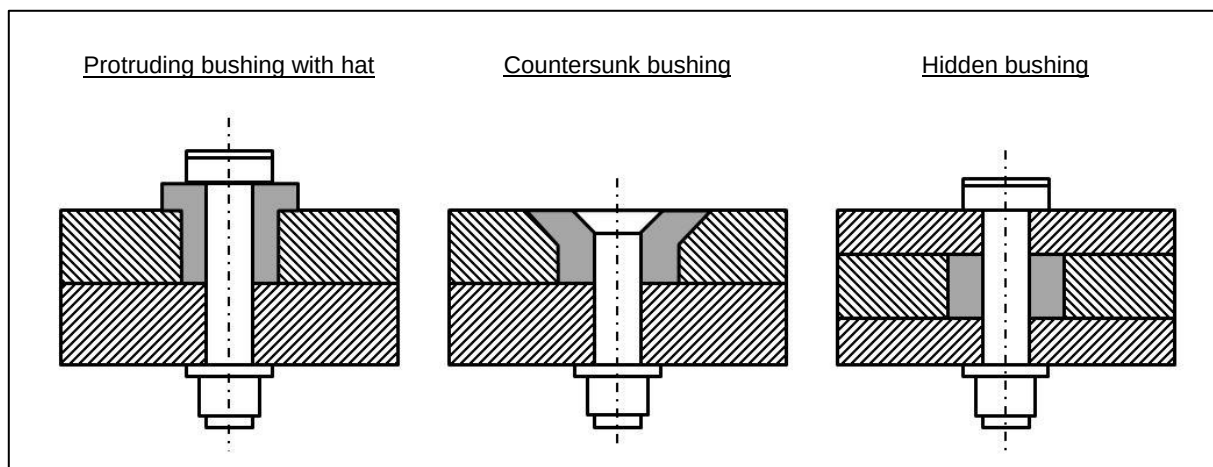


Figure 12. Types of bushings installed at analyzed fastener.

Allowable values A , B and C can be obtained according to the laminate material and bolted joint parameters (laminate thickness, fastener pitch, fastener diameter, bushing type, hole diameter, etc.), taking into account the bushing installation details.

Notice that this method is valid if Airbus DS general design criteria for bushings installation are satisfied and bushing is installed in the composite part according to the same kind of tolerances as the nominal fastener installation at same location.

It should be also remarked that this method does not cover bushing itself strength failure modes.

Tension/Compression By-pass strength - A allowable value calculation:

Allowable value A at installed bushing can be obtained using URCO program according to the following considerations in input data definition:

Fastener diameter = Diameter of the hole where bushing is installed.

Type of joint = Single shear.

Type of fastener = Protruding or Countersunk, the same of bushing geometry, cylindrical hole or countersunk hole, respectively (see Figure 12).

Other input data parameters should be considered as nominal fastener analysis: laminate fibre-layer percentage, thickness, effective width, edge distance, etc.

Tension Through the Hole strength - B allowable value calculation:

Allowable value B at installed bushing can be obtained using URCO program according to the following methodology.

First of all, an auxiliary B_{DS-D} allowable value at bushing must be obtained with URCO according to the following considerations in input data definition:

Fastener diameter = Diameter of the hole where bushing is installed.

Type of joint = Double shear.

Type of fastener = Protruding or Countersunk, the same of bushing geometry, cylindrical hole or countersunk hole, respectively (see Figure 12).

Other input data parameters should be considered as nominal fastener analysis: laminate fibre-layer percentage, thickness, effective width, edge distance, etc.

After that, two other auxiliary allowable values, associated with the installed fastener (not with bushing), B_{DS-d} and B_{SS-d} must be obtained with URCO according to the following considerations in input data definition:

Fastener diameter = Diameter of fastener installed at bushing (not bushing diameter).

Type of joint = Double shear for B_{DS-d} calculation and Single shear for B_{SS-d} calculation.

Type of fastener = Protruding or Countersunk, the same of bushing geometry, cylindrical hole or countersunk hole, respectively (see Figure 12).

Other input data parameters should be considered as nominal fastener analysis: laminate fibre-layer percentage, thickness, effective width, edge distance, etc.

Once B_{DS-d} and B_{SS-d} values have been obtained, the joint eccentricity effect for single shear joints, KDF_{ecc} , can be estimated for B allowable with the following expression:

$$KDF_{ecc} = \frac{B_{SS-d}}{B_{DS-d}}$$

Then, the allowable value B at installed bushing will be:

$$\text{Double shear joints} \rightarrow B = B_{DS-D}$$

$$\text{Single shear joints} \rightarrow B = B_{DS-D} \cdot KDF_{ecc}$$

And tension through the hole stress must be calculated as:

$$\sigma_{TH} = \frac{P}{t \cdot D}$$

Where P is the shear load reacted by the fastener at bushing, t is the laminate thickness and D is the diameter of the hole where the bushing is installed.

Bearing strength - C allowable value calculation:

Allowable value C_{URCO} at installed bushing can be obtained using URCO program according to the following considerations in input data definition, if bushing is not located close to a true edge of the part:

Fastener diameter = Diameter of fastener installed inside the bushing (not bushing diameter).

Type of joint = Single shear.

Type of fastener = Protruding or Countersunk, the same of bushing geometry, cylindrical hole or countersunk hole, respectively (see Figure 12).

Other input data parameters should be considered as nominal fastener analysis: laminate fibre-layer percentage, thickness, effective width, edge distance, etc.

And bearing stress must be calculated as:

$$\sigma_{br} = \frac{P}{t \cdot D}$$

Where P is the shear load reacted by the fastener at bushing, t is the laminate thickness and D is the diameter of the hole where the bushing is installed.

It is recognized that for some bush repair configurations and during concessions, lower edge distance to bolt diameter e/D ratios can occur so, although it is outside the recommended design practices or design principles of aircrafts, If bushing is located close to a true edge of the composite part and all e/D ratios around bushing are equal or higher than 1.0 (being e the edge distance and D the diameter of the bushing), the pure bearing allowable value C at installed bushing can be also calculated with above mentioned considerations but also applying to C_{URCO} value an additional edge distance effect knock-down factor KDF_{edge} . In absence of an internal dedicated test data, it is noted that Airbus Commercial proposes a method to obtain this knock-down factor in sub-chapters 12.3.1 and 12.4 of Ref. 6 (*Technical Report V020RP0818061 issue 6.0 'Static Method for Sizing of Composite Bolted Joints'*), for edge distance ratios e/D over 1.0.

Notice that in no circumstances the countersunk head of a bushing (if applies) shall overlap or be too closely approach the edge of the composite part. Such configurations are not covered by the current method and the fasteners shall be excluded from the load transfer calculation. The design shall ensure that the load can be safely distributed between the remaining fasteners of the joint.

A.7 ALLOWABLE VALUES DIRECTION SELECTION CRITERION

According to the most common configurations, loads reacted by the analysed fastener may be applied in any known orientation θ , which does not usually match with any one of the main directions of the laminate lay-up sequence definition (0° , $+45^\circ$, -45° and 90°); therefore, in order to analyse those design configurations by means of URCO allowable values, it should be firstly selected which laminate's main direction is closer to the orientation of applied load.

The generally accepted approach is selecting the laminate's main direction for URCO allowable values calculation (0° , $+45^\circ$, -45° or 90°) attending to the polar region around fastener which includes the orientation of applied load, θ . Figure 13 defines the four possible regions A, B, C and D, one per each main direction so, according to this mapping, the applicable URCO allowable values should be obtained as follows:

if $-22.5^\circ < \theta < 22.5^\circ$ (Region A) → Allowable values at 0° direction must be used.

if $22.5^\circ < \theta < 67.5^\circ$ (Region B) → Allowable values at $+45^\circ$ direction must be used.

if $67.5^\circ < \theta < -67.5^\circ$ (Region C) → Allowable values at 90° direction must be used.

if $-67.5^\circ < \theta < -22.5^\circ$ (Region D) → Allowable values at -45° direction must be used.

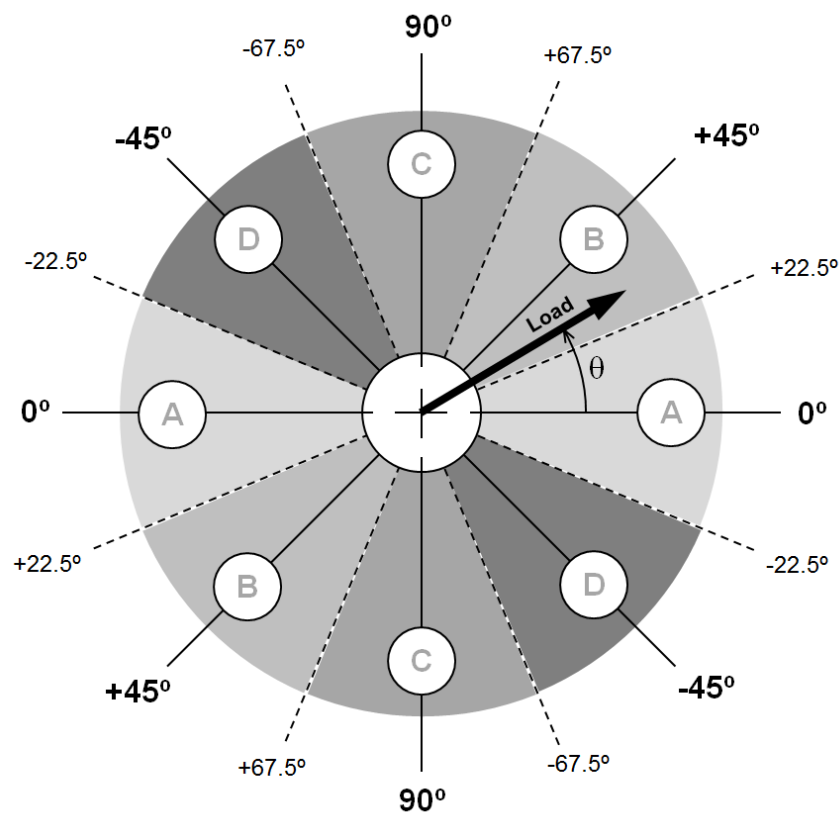


Figure 13. Regions around fastener for URCO allowable values direction selection.

The previously described methodology can be followed to select the most accurate corresponding allowable values for both Bearing and By-pass/TTH reserve factors calculations but, due to the anisotropic behaviour of usually defined laminates, potential relevant errors could be made following this methodology if applied load orientation θ is very near to the angle associated to a frontier between two adjacent regions (intermediate frontiers e.g. $\theta = -22.5^\circ$ or $\theta = 67.5^\circ$, see Figure 13). In case of failure modes dominated by mechanical properties of fibres, such as laminate net cross-section strength at a hole, neglecting these potential deviations are traditionally considered acceptable due to definition of allowable values A and B only make sense if they are aligned with laminate main orientations 0° , $\pm 45^\circ$, and 90° , not with any intermediate one so, no additional post-process of URCO allowable values A and B is usually required for those analyses. In the other hand, in case of failure modes dominated by mechanical properties of resin, such as laminate bearing strength at a hole, a good approach to obtain enough accurate bearing reserve factors in intermediate θ directions is estimating the final bearing allowable value C by linear interpolation between the C values associated to the two adjacent regions (according to the URCO databases, allowable value C is linearly dependant with lay-up fibre percentages).

Nevertheless, in spite of all above mentioned, if a structural critical location is being analysed for an intermediate θ load angle (θ close to -22.5° or 67.5° and far from main directions 0° , $\pm 45^\circ$, 90°), it is highly recommended to perform the analysis by using a conservative envelope of the minimum allowable values of the two adjacent regions, in order to minimize as much as possible the potential error caused by the certain losses of laminate strength properties at θ direction.

A.8 ANALYSIS OF MULTI-AXIALLY LOADED JOINTS

There are several design configurations where laminates should withstand multi-axial load states at installed fasteners, such as fasteners belonging to continuous attachments (e.g. frame to skin joint) or attachments with additional external loads (e.g. pressurization or thermal loads), etc.. In those situations, the orientation of critical shear load reacted at the fastener and the orientation of critical by-pass load may not be the same (angles θ and φ in scheme shown in Figure 14, where ε_x , ε_y and γ_{xy} correspond to far field axial strains applied to the analysed laminate) so, the following conservative approach is suggested in order to calculate the joint's By-pass/TTH interaction reserve factor using URCO allowable values (Bearing reserve factor can be obtained as usual way).

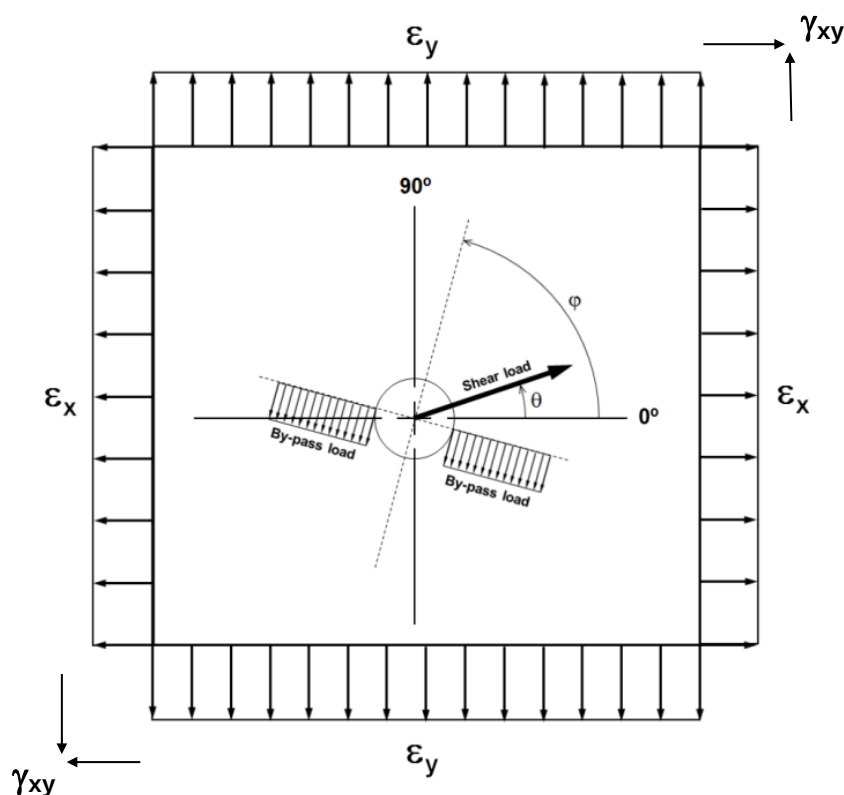


Figure 14. Multi-axially loaded fastener scheme.

First of all, the maximum by-pass strain value and its orientation angle φ for the analysed load case must be obtained and compared with laminate's URCO allowable value A at the same orientation of this strain. The maximum by-pass strain is generally considered as the maximum applied one in the laminate's main directions 0° , 45° and 90° , which are the orientations aligned with fibres; therefore, the direction of allowable value A to be used in By-pass/TTH reserve factor calculation should be the one associated to the direction of the selected maximum by-pass strain, then, angle φ value may be equal to 0° , $+45^\circ$, -45° or 90° (See sub-chapter A.7 for joint allowable value A criterion of selection attending to the applied loads orientation).

Secondly, for the same load case, the maximum bearing stress reacted by the fastener and the orientation angle θ of its associated shear load must be also obtained and compared with laminate's

URCO allowable value B at the same orientation of this stress (See sub-chapter A.7 for joint allowable value B criterion of selection attending to the applied bearing stress orientation).

Then, according to all herein above, the By-pass/Tension-through-the-hole interaction reserve factor of the joint can be calculated by the following expression:

$$RF_{Bypass/TTH} = \frac{1}{\frac{\sigma_{\theta}}{B_{\theta}} + \frac{\varepsilon_{\varphi}}{A_{\varphi}}}$$

Where:

- | | |
|-------------------------|--|
| σ_{θ} | Maximum bearing stress reacted by fastener for the analysed load case [MPa] |
| ε_{φ} | Maximum by-pass strain around fastener for the analysed load case [$\mu\varepsilon$] |
| B_{θ} | Selected URCO allowable value B for direction θ of max. applied bearing stress [MPa] |
| A_{φ} | Selected URCO allowable value A for direction φ of maximum applied by-pass strain [$\mu\varepsilon$] |